ESL Brains

What are the odds?

1. **Say how much you agree with each sentence and provide more details.**
   - Small coincidences happen every day if we pay enough attention.
   - Coincidences often have a great impact on your life.
   - If people knew more about probability they wouldn't believe in coincidences.

2. **Choose a group in the boxes that you belong to and discuss the questions.**

your family | your colleagues | your friends

your neighbours | your classmates

   - Do you know if you share
     - a birthday?
     - a birth month?
     - a birth year?
   - How many people in this group do you think share a birthday? Why do you think so?

3. **You are going to watch a video about the phenomenon called 'the birthday paradox'. Before you watch, answer the question in the box.**

> How big do you think a group of people would have to be before there's more than a 50% chance that two people in the group have the same birthday?

4. **Watch the first part of the video [https://youtu.be/KtT_cgMzHx8] (to 00:56) and check your answer.**

5. **Discuss the questions.**
   - Are you surprised by the answer in ex. 4? Why/Why not?
   - How can this number be explained? What could be the logic behind it?

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6. **Look at the sentences and think of a word or a number to complete each gap. Then, watch the rest of the video (00:56–04:50) and check your answers.**

   A. Mathematicians use a field called combinatorics to calculate the _____ of different combinations.

   B. Instead of calculating the probability of a birthday match directly, it's easier to calculate the probability that everyone's birthday is _____ .

   C. This works because the probability of a match and the probability of no match must equal _____ %.

   D. If Person A already has one of the 365 possible birthdays, Person B has _____ possible birthdays that don't match A's.

   E. To calculate the probability that all 23 people have different birthdays, all these individual probabilities are _____ together.

   F. The main reason this probability of two people sharing a birthday is so high is the surprisingly large number of possible _____ within a group.

   G. With _____ people, the probability of a shared birthday rises to over 99.9%.

7. **Discuss the questions.**

   - What do you think about the birthday paradox?
   - Why do you think so many people get this problem wrong?
   - Has the video changed the way you think about coincidences? Why/Why not?
   - Do you think considering probability can affect important decisions? You may consider the following:
     - boarding a plane
     - practising extreme sports
     - getting married
     - going to university
     - starting your own business
   - Has anything happened in your life that you would describe as a coincidence? Provide more details.

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8. Read the sentences about other examples of coincidences and match the phrases in bold with their definitions (1–8).

A. I was thinking about a friend I hadn't spoken to in years, and five minutes later she messaged me. **What are the odds?**

B. I had **a nagging feeling** that I'd forgotten something, so I checked my bag one last time and realized my passport was still on the kitchen table.

C. I almost cancelled the trip because of the weather forecast, but something told me conditions would improve, so I decided to **go with my gut** and went anyway.

D. We booked the same tiny hotel in Portugal for exactly the same week, without ever discussing our plans. It was **sheer coincidence**.

E. Everyone was shocked when the two business partners fell out, but I could **see it coming** from the way they'd been arguing for months.

F. I kept seeing the same man on different trains, but having similar work schedules is a **plausible** explanation.

G. My sister thinks seeing the same number everywhere is some kind of sign. I think she's **reading too much into** it.

H. **In hindsight**, it made sense that we were reading the same book as it had been all over social media that month.

1. a worry that stays in your mind and is difficult to ignore
2. finding more meaning or importance in something than it probably has
3. looking back at a past situation with the knowledge you have now
4. realize or expect that something is going to happen before it happens
5. seeming reasonable and likely to be true or possible
6. something that happens completely by chance
7. trust your feelings or instincts when making a decision
8. used to express surprise that something unlikely has happened or is happening

9. Look at the sentences in ex. 8 again and say if something similar has ever happened to you or someone you know. Provide more details.

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10. Complete each gap with one word. The first letter has been provided. Then, choose four points to talk about.

A. a time when you went with your **g**_____ but later regretted it

B. a situation when you had a **n**_____ feeling that something was wrong

C. a time when you thought: "What are the **o**_____ ?"

D. a situation when the most **p**_____ explanation turned out to be completely wrong

E. something you wish you had done differently in **h**_____

F. a time when you read too much **i**_____ someone's words or actions

G. a **s**_____ coincidence that seemed almost impossible

H. a time when you saw a problem **c**_____ before anyone else did

11. Choose four examples of coincidences in the boxes and do the tasks below. Use the vocabulary in the box.

You wear the same colour or a very similar outfit as someone else at work without planning it.	You predict that two friends will break up. They do two weeks later.
You think about pizza at lunchtime and then see three pizza ads online.	You have a bad feeling before getting on a plane. The flight experiences severe turbulence.
Three members of the same family independently buy each other the same present for Christmas.	You meet someone at a conference who went to your university.
You discover that three people in your English class have the same name.	You meet a stranger on holiday abroad and discover that you live in the same neighbourhood.

A. Say how you would react in the situation.

B. Discuss whether you consider the situation to be a coincidence or whether it has a logical explanation.

go with your gut	read too much into
in hindsight	see something coming
nagging feeling	sheer coincidence
plausible	What are the odds?

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12. Read about famous coincidences in history and discuss the questions below.

In 1898 Morgan Robertson published a novella called *Futility*, which imagined the sinking of a British ocean liner named Titan. Like the Titanic, it hit an iceberg in April, was considered 'unsinkable', and didn't have enough lifeboats. The Titanic's real sinking happened 14 years later.

The Jim twins – identical twins separated at birth – were raised apart in Ohio, yet their lives turned out remarkably similar. Both were named James, both married women named Linda, and both had sons named James Allan. The twins also chose similar careers and became law enforcement officers. At age 39, they finally met.

- Were you familiar with either story? Which of them do you find more remarkable? Why?
- Do you think what happened in both stories are sheer coincidences, or might there be a plausible explanation for either of them?
- Do you know of any stories similar to *Futility*, in which someone seemed to see something coming before it happened? Share more details.
- If the story wasn't about identical twins, would people still read so much into the similarities in their lives? Why/Why not?



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